Diego: Here only 2 pages with problem description

Objective

The plan is to place a few applications on multiple servers, segregated by how critical they are to the infrastructure. Each server must provide enough CPU, RAM, and if required GPU to satisfy runtime needs.

Problems

Match each application's requirements with an appropriate server while respecting hardware limits, criticality, and basic security or connectivity.

Domains

Applications, Servers, and Network (front/back segments).

Relation

ApplicationMM and ServerMM and NetworkMM
↓↓↓↓↓↓↓↓↓↓↓↓ **M2M** ↓↓↓↓↓↓↓↓↓↓↓↓
PlacementMM
↓↓↓↓↓↓↓↓↓↓↓↓ **M2T** ↓↓↓↓↓↓↓↓↓↓↓↓
Deployment.yml and firewalls rules

Models

Metamodels

1. Application MM (represent what we need to run)
   Fields: name, cpu, ram, gpu, critical, network

2. Server MM (represent where we can run)
   Fields: name, cpuCoresTotal, ramTotal, gpu, networks[]

3. Network MM (represent how things connect)
   Fields: name, allowed_pairs[]

4. Placement MM (represent the solution for the problem we compute)
   Fields: app, server, notes[]

Example models:
Applications.model (ApplicationMM)

- Web, cpu=2, ram=4GB, gpu=0, critical=1, network=front
- ServerAPI, cpu=4, ram=16GB, gpu=0, critical=1, network=back
- Database, cpu=4, ram=16GB, gpu=0, critical=2, network=back

Servers.model (ServerMM)
- Server1: cpuCoresTotal=4, ramTotal=16GB, gpu=0, networks=[front, back]
- Server2: cpuCoresTotal=8, ramTotal=16GB, gpu=0, networks=[front, back]
- Server3: cpuCoresTotal=16, ramTotal=32GB, gpu=0, networks=[front, back]

Network.model (NetworkMM)

- Networks: front, back
- allowed_pairs: Web -> ServerAPI, ServerAPI -> Database
- not allowed: Web -> Database

Tool integration

Produce

Forms from Application.model. CMDB "Configuration Management Database" and export from Server.model. JSON from Network.model.

Consume

Deployment.yaml to docker compose and Firewall.rules to firewall manager.

Model creation and updates

The application, server and network models will be created manually from the specified requirements listed. Whenever changes are conducted to the models it triggers T1 to T2 transformation and updates Placement.model and then creates M2T automatically.

Model validation

App.cpu requirement has to be lower or equal than server.cpuCoresTotal, app.ram lower or equal to server.ramTotal and app.gpu lower or equal than server.gpu. Eventually app.network has to match server.networks. Applications with critical equal to 1 or higher should not share a server. The summary of assigned applications CPU/RAM AND GPU does not exceed totals and traffic respect allowed_pairs i.e WEB to database denial.